



Fig. 4: EFLX DSP cores accelerate signal processing

LP/ULP process. Evaluation and prototyping of a single-core EFLX design gives customers the experience and insight to build flexibility into their RTL designs, even if the target application requires a larger EFLX array. EFLX cores available are TSMC 28HPM/C EFLX-2.5K core (either logic or DSP version) and TSMC 40ULP/LP EFLX-100 core (either logic or DSP version).

Benefits

EFLX cores allow a single SoC design to adapt to multiple requirements quickly using RTL re-configurability instead of locking in an SoC design with hard-wired logic, or trapping something deeply proprietary for a specific customer. These cores also allow field reconfiguration, which is essential for long-lifecycle devices.

EFLX makes the toolkit available to designers for post-production flexibility. By adding BRAM and DSP block RAM architecture, one can provide exactly the type and amount of memory an application requires. This flexibility is accomplished by inserting BRAM between EFLX logic cores, which tile together to make an array controlling these with otherwise unused inward-facing inter-tile I/Os.

External RAM is mapped and compiled onto the array. EFLX cores can be used to upgrade I/O protocols, change encryption algorithms to improve security, enable elements of software-defined radio or accelerate data centre algorithms like search.

Since EFLX core tiles are independent, these can be spaced apart, leaving room for external RAM integration between tiles. Semiconductor designers would have the ability

to quickly upgrade their designs, equipment manufacturers would be able to modify product lines as required and end users would be able to optimise performance of their systems to address their own unique use cases and extend their product lifecycles. EFLX arrays could really help system-level SoC architecture.

Conventional IP model

EFLX supplies a full set of tools for populating the FPGA design. Adding a 2500-LUT core to a design should add only 15 cents to the total manufacturing cost of your device.

Benefits are board complexity, form factor, power consumption and performance. It also eliminates the need for all those external I/Os that would have connected your custom chip to the external FPGA and all PCB traces that would have carried those signals.

Fabric is the key to both performance and scalability. Addition of low-latency memory and signal-processing capabilities significantly increase the range of applications addressed by EFLX embedded FPGA-in-SoC architecture.

EFLX compiler

EFLX compiler with full timing files is available under a simple software-evaluation licence. This would enable you to determine area, performance, architecture and floor planning using EFLX-reconfigurable RTL. If you wish to proceed to design and build an evaluation chip, it would provide a fast-track licence for a single-core EFLX array for US\$ 50,000. The licence would include full Verilog model, GDS-II, CDL,

integration guidelines, EFLX compiler licence with full timing files and bit-stream generation.

Integration assistance will also be provided. To upgrade an EFLX FPGA, you would need to upgrade its RTL code, run it through Simplify Tool to produce an electronic design interchange format netlist and then use EFLX compiler to reconfigure the internal FPGA on the SoC. For such customers, the company has written its own compiler that allows easy upgrades in the field. The business plan and more-efficient FPGA architecture also reduce the number of metallisation layers to lower cost per gate.

Flow

- Make input RTL to see resources required, that is, LUTs, DSP blocks and RAM
- Configure EFLX array
- Select EFLX array size, optional DSP, type and amount of RAM, clocks and I/O configuration to connect EFLX array with the rest of the SoC
- Input RTL to see the worst-case path and frequency for target EFLX array/node
- Generate bit file that programs EFLX array in the SoC to execute RTL

The basic building block for implementing DSP functions is a pre-adder MAC EFLX logic core that incorporates 40 MACs with 22-bit inputs and 48-bit accumulation. The MACs can be combined for double the precision and pipelined for high throughput. These can also be used as complex-number MACs for certain DSP algorithms.

Trend setters

The three main market segments for EFLX core are networking, wireless/digital front-ends and micro-controllers (MCUs). With the sheer number of I/Os EFLX tile has, it can implement 256b or even 512b busses and put logic on the pins that need it.

Another useful feature of program-